

## Every origin is first normalised to a [time, value] pair

None → [None, None]

0.5 → [0.5, None]

"molasses" → ["molasses", None]

[a, b] → [a, b]

Each slot is then completed independently: one left as None takes the caller's default for that slot.

### time slot

*every option names an instant*

|                 |                                                        |
|-----------------|--------------------------------------------------------|
| None            | the caller's chain: anchor → last → 0.0 with a warning |
| a number        | that number                                            |
| "anchor"        | the most recent anchor — an error if there is none     |
| "last"          | the highest time recorded so far                       |
| "variable"      | per variable: its own most recent time                 |
| a variable name | that variable's most recent time                       |
| a context name  | its anchor, or its last row if it has none             |
| anything else   | an error                                               |

### value slot

*only a variable names a quantity*

|                                     |                                                         |
|-------------------------------------|---------------------------------------------------------|
| None                                | the caller's default: absolute — or "variable" for ramp |
| a number                            | that number                                             |
| "variable"                          | per variable: its own last value †                      |
| a variable name                     | that variable's last value †                            |
| "anchor", "last",<br>a context name | refused — each names an instant, not a quantity         |
| anything else                       | an error                                                |

† Bounded by the resolved time origin: the value in effect at the instant the new rows will occupy, not the variable's last value in the timeline as a whole. That is what lets an operation be interwoven and still see the state that precedes it.